

NIDPay — Team Task Distribution

BoU@60 Hackathon | React · Spring Boot · PostgreSQL · MTN & Airtel APIs

1. Project Stack

- **Frontend:** React (TypeScript) + TailwindCSS — wallet, QR, merchant dashboard.
- **Backend:** Spring Boot (Java 17) — REST APIs, JWT auth, transactional services.
- **Database:** PostgreSQL 15 — ACID transactions, RLS, stored procedures, Flyway migrations.
- **Payments:** MTN MoMo API (Collection & Disbursement) and Airtel Money Open API.
- **Infra:** Docker, GitHub Actions CI/CD, deployed to a Ugandan-region VPS.

2. Team Roles & Responsibilities (6 members)

Member	Role	Responsibilities	Tech Focus
Steven Kiwanuka Kafeero	Team Lead / Solutions Architect	Overall architecture, sprint planning, code reviews, security strategy (PIN hashing, RLS, audit), DevOps pipeline, BoU integration liaison.	Architecture, Security, DevOps
Kalema Kosea	Backend Lead — Spring Boot Core	Core Spring Boot services: User, Wallet, Transaction, PIN. JPA entities, repositories, atomic debit/credit service, JWT auth.	Spring Boot, JPA, PostgreSQL
Kalemera Collins	Backend — Payments & API Integration	MTN MoMo (Collection & Disbursement) and Airtel Money API integration. Webhook handlers, signature verification, retry/idempotency, reconciliation.	Spring Boot, MTN/Airtel, Webhooks
Mugabi William	Backend — Database & Business Logic	PostgreSQL schema design, Flyway migrations, stored procedures, indexing, fraud-rule engine, transaction limits, KYC validation.	PostgreSQL, SQL, Business rules
Musigazi Clinton	Frontend Lead — React	React architecture: routing, state, wallet UI, QR scanner, merchant dashboard, transaction history, PIN entry UX, responsive design.	React, TailwindCSS, REST
Mawerere Jordan Mark	QA, Testing & Documentation	JUnit unit tests, integration tests, Postman/E2E flows for MTN & Airtel sandbox, load testing, user docs, demo script & pitch deck.	JUnit, Postman, JMeter

3. Module Ownership Map

- Auth & Users → Kosea (backend) + Clinton (UI)
- Wallet & Balances → Kosea + Mugabi (DB) + Clinton (UI)
- PIN Authorization → Steven (security design) + Kosea (impl) + Clinton (entry UX)
- MTN MoMo top-up & withdrawal → Collins
- Airtel Money top-up & withdrawal → Collins
- Merchant payments & QR → Kosea (API) + Clinton (scanner UI)
- Fraud limits & rules → Mugabi

- Admin / audit dashboard → Clinton + Mugabi (queries)
- CI/CD, deployment, env secrets → Steven
- Testing, docs, pitch → Jordan

4. MTN & Airtel Integration — Split

Collins (owner)

- MTN MoMo: Sandbox API user/key provisioning, OAuth token caching, RequestToPay (collection), Transfer (disbursement), status polling.
- Airtel Money: OAuth client_credentials flow, Collections (USSD push), Disbursements, transaction enquiry.
- Webhook endpoints under /api/webhooks/{mtn,airtel} with signature verification and idempotency keys.

Mugabi (support)

- Reconciliation job (cron) that matches webhook events with the wallet ledger; flags mismatches.

Steven (support)

- Secret storage (MTN subscription keys, Airtel client_id/secret), rotation policy, sandbox→production switch.

5. Timeline (4 weeks)

Phase	Focus	Deliverables	Owners
Week 1	Setup & Design	Repo, CI/CD, DB schema, API contracts, UI wireframes	All
Week 2	Core Build	Auth, Wallet, Transactions, React shell, MoMo sandbox connect	Backend + Frontend
Week 3	Integration	MTN + Airtel live flows, QR payments, merchant dashboard	Collins, Clinton, Kosea
Week 4	QA & Demo	Testing, fixes, documentation, pitch rehearsal	Jordan + All

6. Daily Working Rhythm

- Daily 15-min standup (what I did / blockers / today).
- Feature branches → PR → 1 reviewer (Steven for backend, Clinton for frontend).
- Shared Postman workspace for MTN/Airtel sandbox calls.
- Weekly demo every Friday — Jordan runs the test checklist.

7. Definition of Done (per feature)

- Code merged to main behind a feature flag.
- Unit tests + at least one integration test passing in CI.
- API documented in the Postman collection.
- UI screen reviewed on mobile + desktop.
- Logged in the audit table where money moves.

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